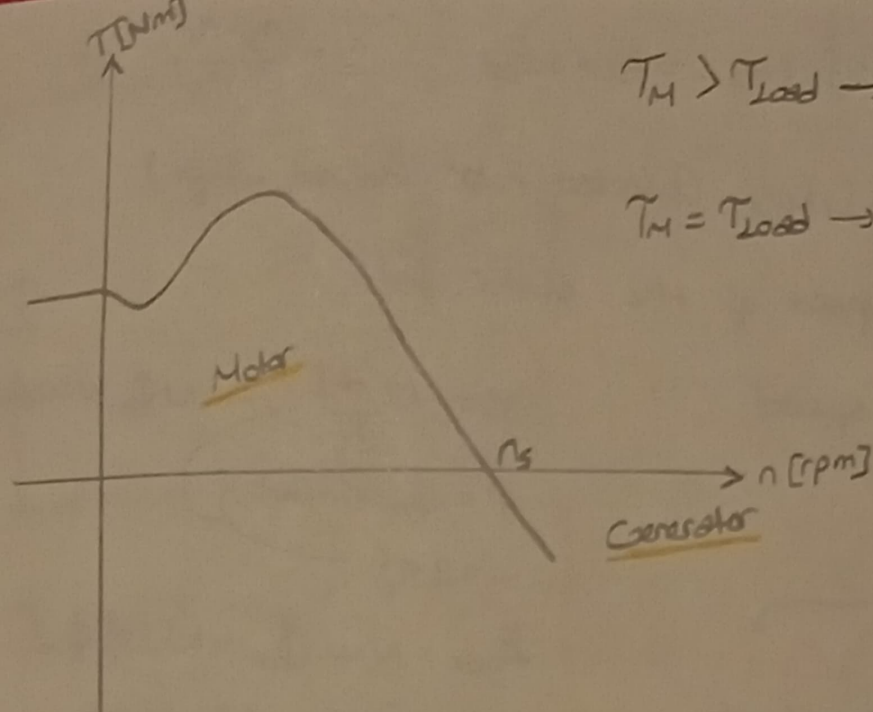


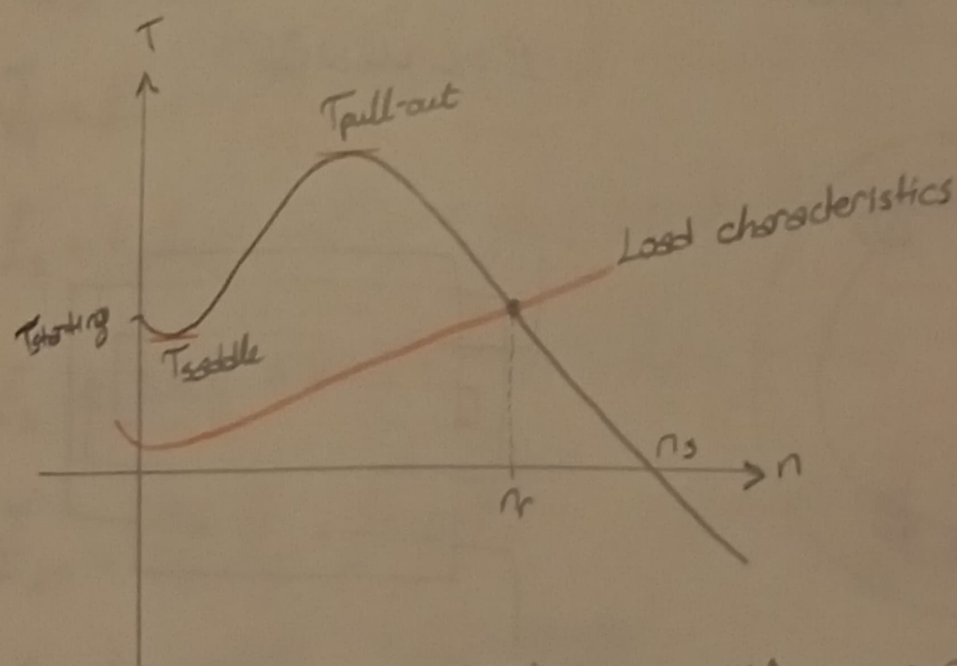


$$T_M > T_{Load} \rightarrow \frac{dw}{dt} > 0 \text{ accelerates}$$

$$T_M = T_{Load} \rightarrow \frac{dw}{dt} = 0 \text{ constant speed}$$

$$T_M - T_{Load} = J \frac{dw}{dt}$$

Inertia



$$e = -N \frac{d\Phi}{dt} = -N \frac{d(BA)}{dt} = -NB \frac{dA}{dt} - NA \frac{dB}{dt}$$

$-NB \frac{dA}{dt}$ → stator magnetic field
ve rotor begin turn

$$E_1 = 4.44 N_1 \Phi_m f_1 k_{w1}$$

winding constant
0.95-0.99

$$E_2 = 4.44 N_2 \Phi_m f_2 k_{w2}$$

$$E_2 = L_{LL} N_2 \Phi_m s f_1 k w_2 \rightarrow E_2 = s E_{20} \quad s = \frac{n_s - n_r}{n_s}$$

$$E_{20} = L_{LL} N_2 \Phi_m f_1 k w_2 \quad (\text{blocked rotor induced voltage})$$

n_s : synchronous speed of the stator field

n_r : actual rotor speed

AC Machines

Induction

- Squirrel cage
- Wound rotor

Synchronous

- Salient pole
- Cylindrical rotor

$$e = -N \frac{d\Phi}{dt} = -N \Phi_m \omega \cos(\omega t)$$

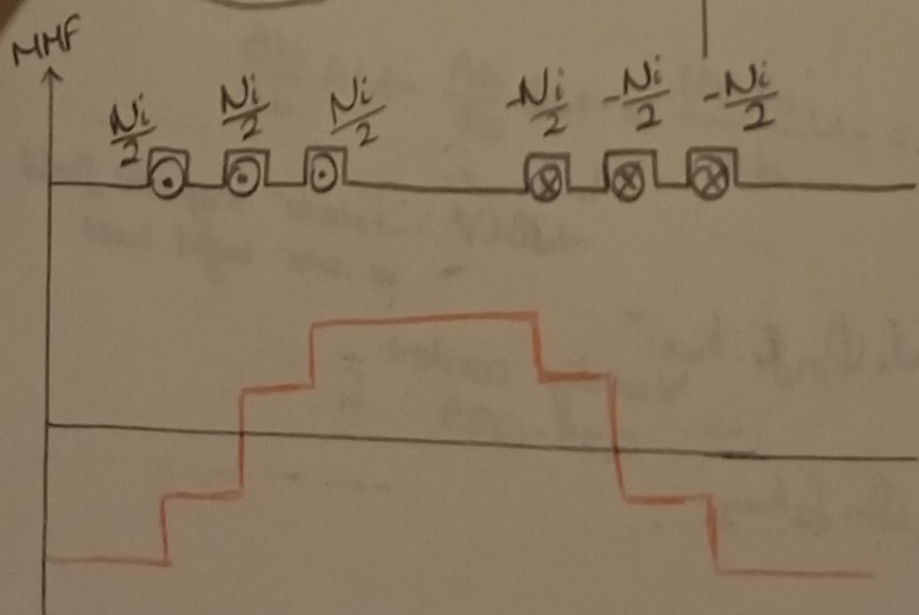
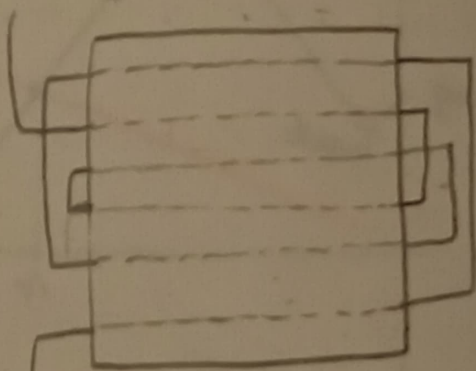
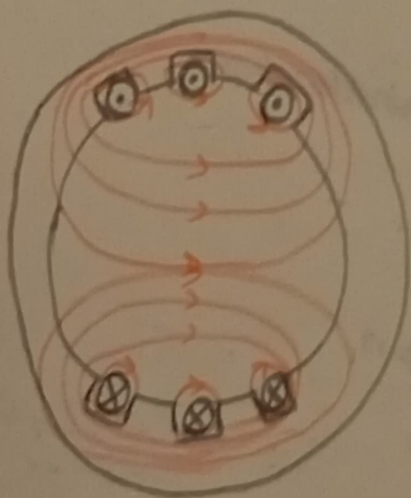
$$\Phi = \Phi_m \sin(\omega t)$$

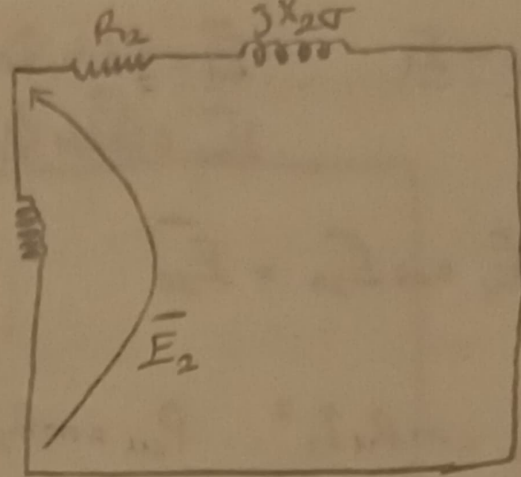
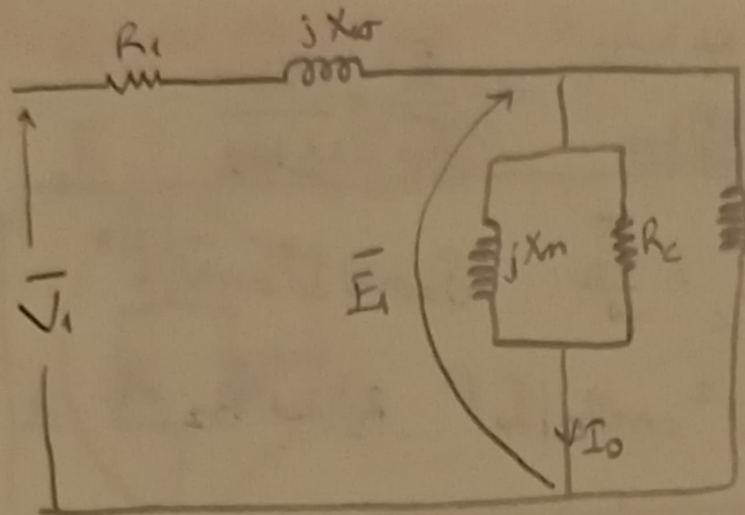
$$\omega = 2\pi f$$

$$E_{\max} = N \omega \Phi_m = N 2\pi f \Phi_m$$

$$E_{\text{rms}} = \frac{E_{\max}}{\sqrt{2}} = \frac{2\pi}{\sqrt{2}} N f \Phi_m$$

$$E = L_{LL} N \Phi f$$





$$\lambda = N\Phi = L\bar{i}$$

$$X = \omega L$$

$$\Phi_{1\sigma} \rightarrow L_{1\sigma} \rightarrow X_{1\sigma}$$

$$\Phi_m \rightarrow X_m \quad (L_m)$$

$$P_{\text{iron}} = P_{sc} = P_c$$

$P_h + P_e$: hysteresis and eddy currents

$$\bar{V}_1 = \bar{E}_1 + \bar{I}_1 R_1 + \bar{I}_1 jX_{1\sigma} \rightarrow \bar{E}_1 = 4.44 f_1 \Phi_m N_1 k w_1$$

$$\bar{E}_2 = R_2 \bar{I}_2 + jX_{2\sigma} \bar{I}_2 \rightarrow \bar{E}_2 = 4.44 f_2 \Phi_m N_2 k w_2 \quad f_2 = s f_1$$

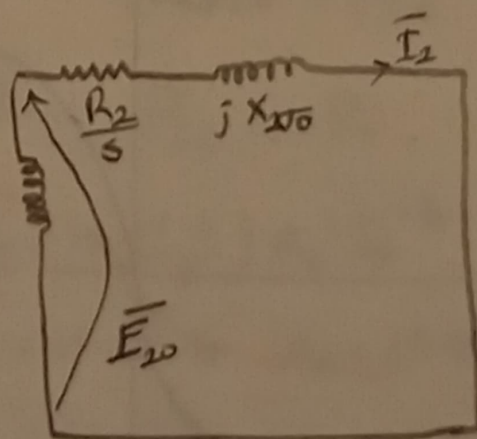
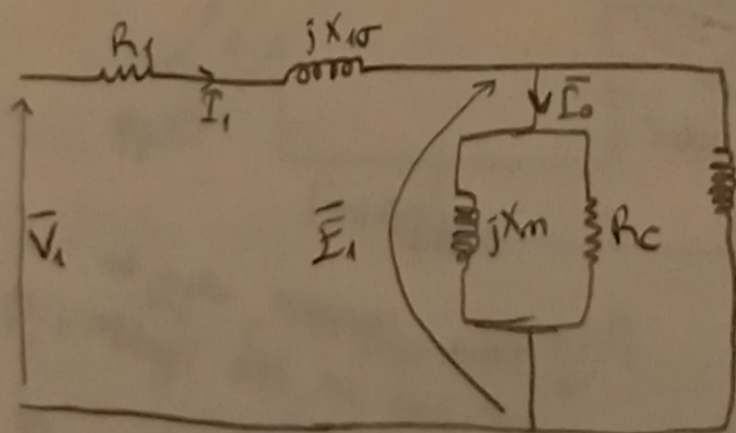
$$\bar{E}_2 = 4.44 f_1 s \Phi_m N_2 k w_2 \rightarrow E_2 = s \cdot E_{20}$$

$$\bar{E}_{20} = \frac{R_2}{s} \bar{I}_2 + jX_{2\sigma} \bar{I}_2$$

$$X_{2\sigma} = \omega L_{2\sigma} = 2\pi f_2 L_{2\sigma}$$

$$X_{2\sigma} = s \underbrace{2\pi f_1 L_{2\sigma}}_{X_{2\sigma 0}}$$

$$\Phi = \Phi_m \sin(\omega t) \quad X_{2\sigma} = s X_{2\sigma 0}$$



$$E_{20}' = E_1 \quad E_1 = L_1 I_1 \Phi_m \delta_1 N_1 k_{w1}$$

$$\bar{E}_{20} = L_1 I_1 \Phi_m \delta_1 N_2 k_{w2}$$

$$\alpha = \frac{\bar{E}_1}{\bar{E}_{20}} = \frac{N_1 k_{w1}}{N_2 k_{w2}}$$

$$\bar{E}_1 = \alpha \bar{E}_{20} = \bar{E}_{20}'$$

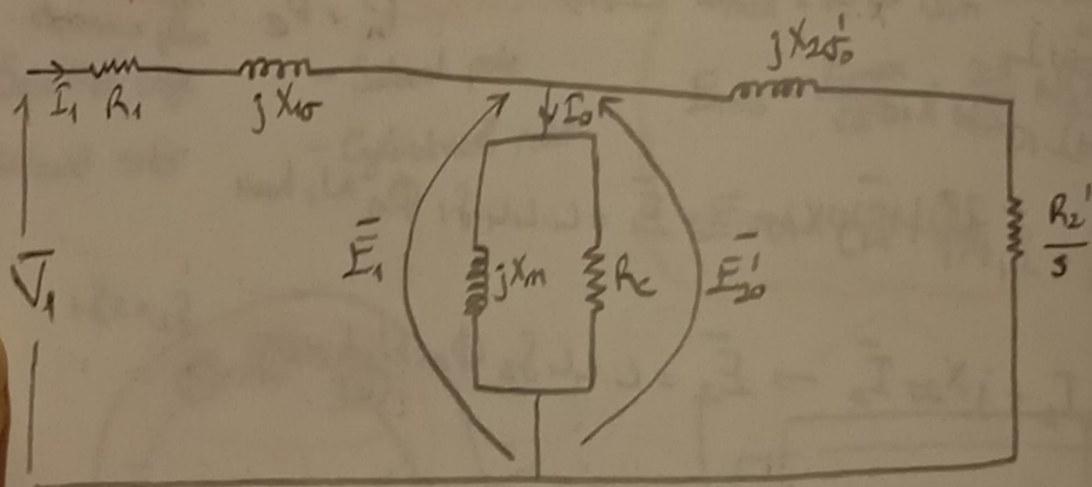
$$\bar{I}_2' = \frac{\bar{I}_2}{\alpha}$$

$$S = m V \bar{I}^* = m \bar{V}' \bar{I}'^*$$

$$P_{cu} = m R_1 \bar{I}_1^2 \quad P_{cu} = m R_2 |\bar{I}_2|^2 = m R_2' |\bar{I}_2'|^2 \quad R_2' = \alpha^2 R_2$$

$$Q_2 = m |\bar{I}_2|^2 X_{20} = m |\bar{I}_2'|^2 X_{20}'$$

T-equivalent circuit of an induction machine

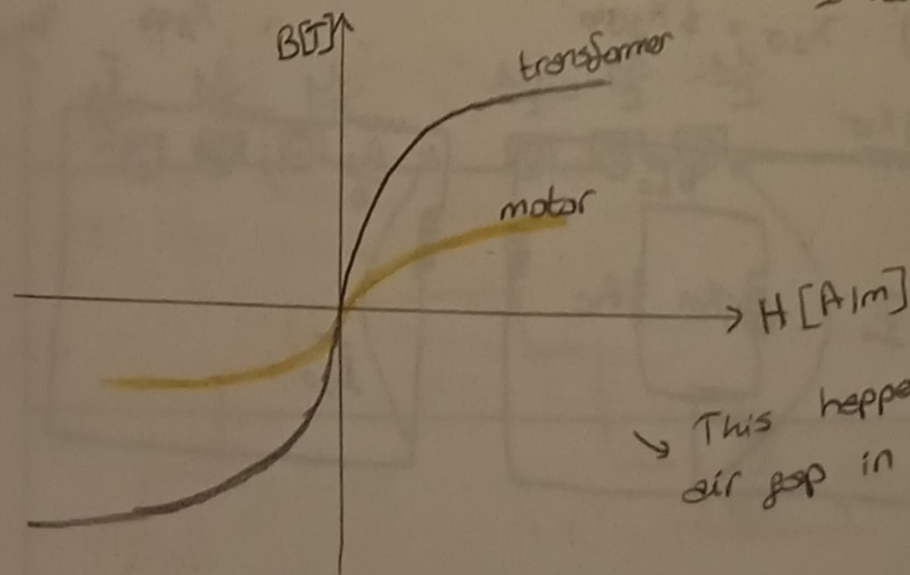


$$\bar{E}_{20}' = \bar{I}_2' \frac{R_2'}{s} + \bar{I}_2' j X_{20}'$$

$$\bar{I}_2' = \frac{\bar{E}_{20}'}{\frac{R_2'}{s} + j X_{20}'} = \frac{s \bar{E}_{20}'}{R_2' + j s X_{20}'}$$

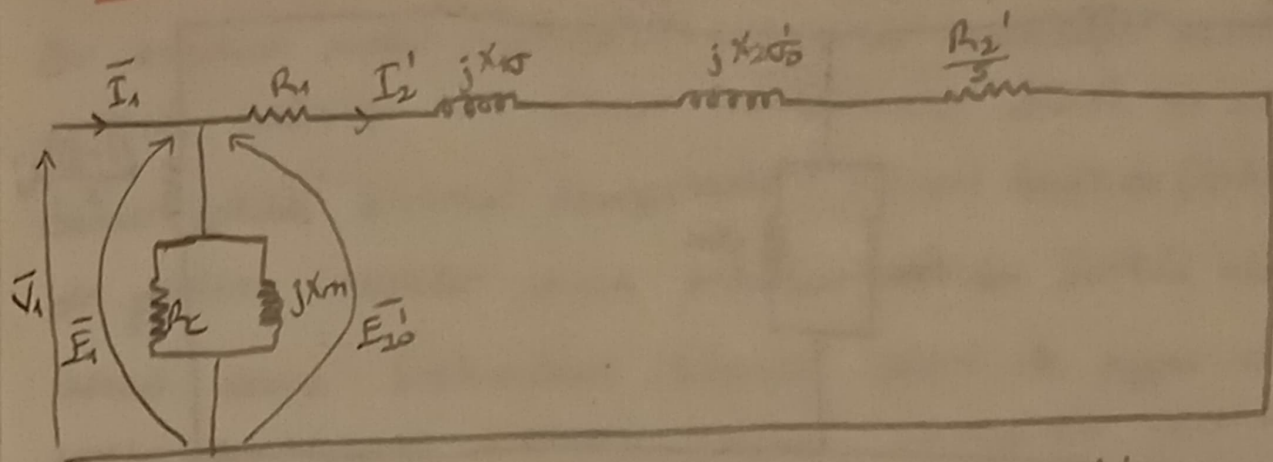
2 marks

$$\bar{I}_1 = \bar{I}_2' + \bar{I}_0 \quad \bar{I}_0 = \bar{I}_{se} + \bar{I}_m$$



→ This happens due to air gap in the motor

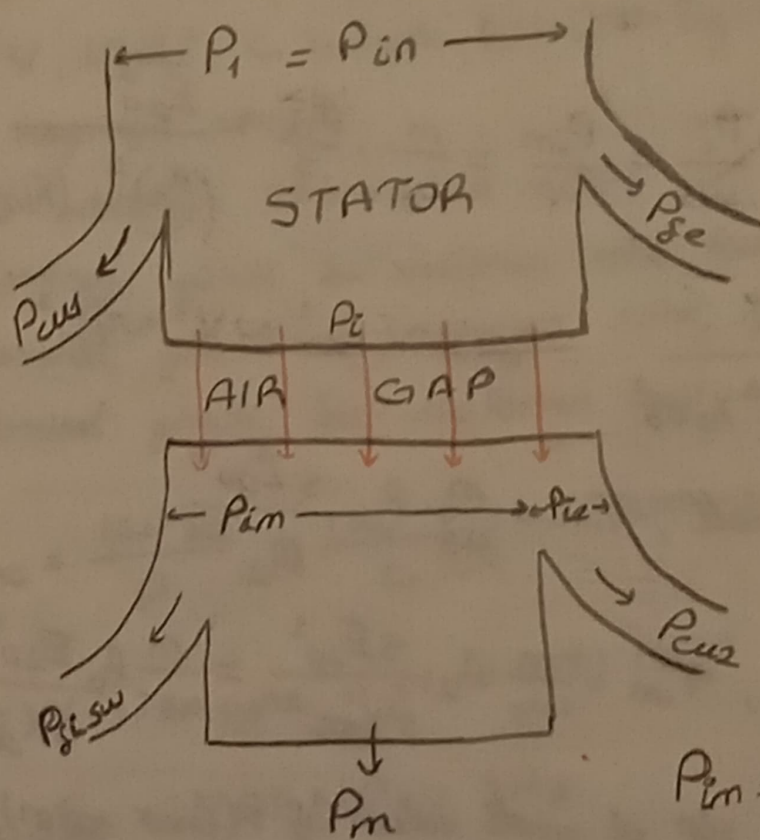
L-equivalent circuit of an induction machine



$$\bar{I}_2' = \frac{\bar{E}_{20}'}{R_1 + \frac{R_2'}{s} + jX_{1\sigma} + jX_{2\sigma 0}'}$$

$$|I_2'| = \frac{|E_{20}'|}{\sqrt{\left(R_1 + \frac{R_2'}{s}\right)^2 + (X_{1\sigma} + X_{2\sigma 0}')^2}}$$

Sonkey Diagram



$$P_1 = P_{\text{input}} = m V_1 I_1 \cos \phi$$

$$P_i = P_{\text{internal}} = P_{\text{air gap}}$$

$$P_i = P_1 - P_{cu1} - P_{se}$$

$$P_{cu1} = m I_1^2 R_1$$

$$P_{sw} = \text{friction and windage losses}$$

$$P_i = P_{im} + P_{ie}$$

$$P_{ie} = s P_i$$

$$P_{im} = m \left(\frac{1-s}{s} \right) R_2' I_2'^2$$

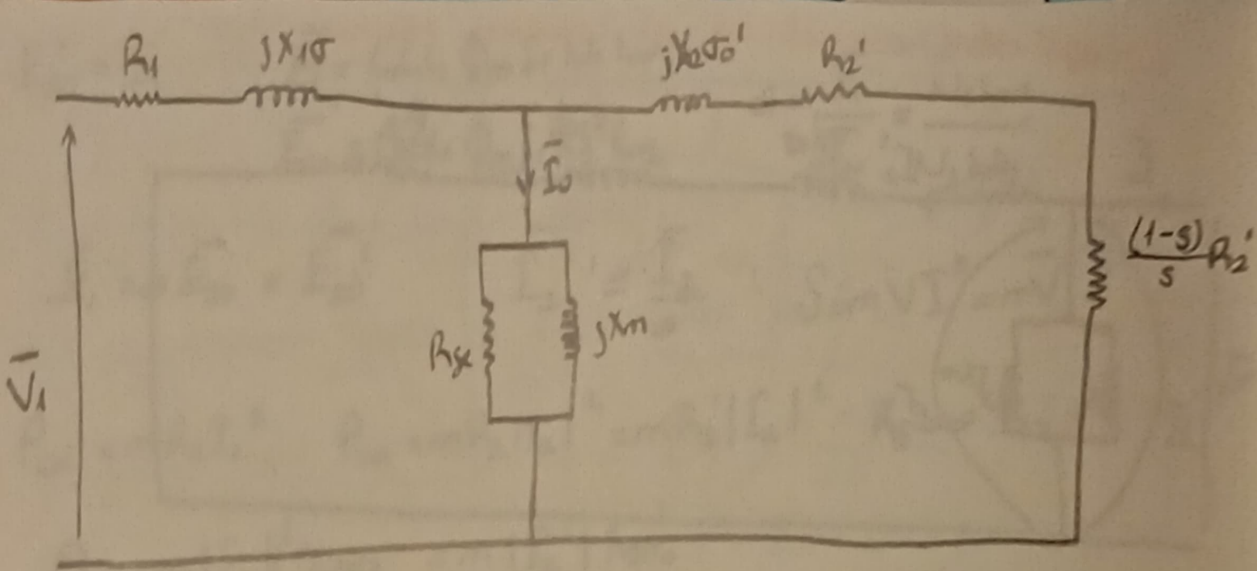
$$n_r = (1-s) n_s \quad n_2 = (1-s) n_1$$

$$P_i = m \frac{R_2'}{s} I_2'^2 \quad P_{im} = (1-s) P_i$$

$$P_{ie} = m R_2' I_2'^2 \quad P_{out} = P_m = P_{im} - P_{sw}$$

$$P_{im} = (1-s) P_i$$

$$P = T \omega$$



$$\frac{R_2'}{s} = R_2' + \frac{(1-s)}{s} R_2' \quad \omega_r = \frac{2\pi n_r}{60} \quad n_r = (1-s)n_s$$

$$\omega_2 = (1-s)\omega_1$$

$$\omega_r = (1-s)\omega_s$$

$$T_{im} = \frac{P_{im}}{\omega_r} \quad \omega_r = \frac{2\pi n_r}{60}$$

$$T_{im} = m \frac{(1-s)}{s} R_2 |I_2|^2 = \frac{m}{\omega_r} \frac{(1-s)}{s} R_2' |I_2'|^2$$

$$= \frac{m}{\omega_s} \frac{R_2}{s} |I_2|^2 = \frac{P_i}{\omega_s} = \frac{P_{im}}{\omega_r} = \frac{m}{\omega_s} \cdot \frac{R_2}{s} \cdot \frac{\bar{E}_{20}^2}{\left(\frac{R_2}{s}\right)^2 + (X_{2\sigma 0})^2}$$

$$T_{im} = \frac{m}{\omega_s} R_2 \frac{s E_{20}^2}{(R_2)^2 + s^2 X_{2\sigma 0}^2}$$

$$T_{im}(s) \sim E_{20}^2 \sim V_1^2 \sim \Phi_m^2$$

$$\text{If } s \text{ significantly small, } T_{im} = \frac{m}{\omega_s} R_2 \frac{s E_{20}^2}{R_2^2}$$

$$\text{If } s \text{ significantly large, } T_{im} = \frac{m}{\omega_s} R_2 \frac{s E_{20}^2}{s^2 X_{2\sigma 0}^2} = \frac{m}{\omega_s} R_2 \frac{E_{20}^2}{s X_{2\sigma 0}^2}$$

Torque is proportional to slip at small values of s (linear region).

But decreases as slip increase further. (If rotor rotate exactly at n_s , there would be no relative motion, no induced current and no torque. Slip allows the rotor to generate torque by creating relative motion)

Asenkron Elektrikli Fikstör

Bir asenkron motor, makinenin rotorunda indüklenen akım ve gerilimle çalıştığı için dönen transformatör olarak da adlandırılır.

Benzer şekilde birincil devre (stator) ikincil devrede (rotorda) bir gerilim indükler ancak transformatörden farklı olarak ikincil devre frekansının birincil devre ile aynı olması gerekmez. Motorun rotoru hareket etmeyecek şekilde kilitli tutulursa rotor ve stator aynı frekansa sahip olacaktır.

$$n_m = 0 \text{ rpm'de rotor frekansı } f_r = f_e \quad \text{→ sistem frekansı}$$

$$\text{arbitrary } f_r = s f_e = \frac{p}{120} (n_{\text{sync}} - n_m) \quad s = 1$$

Örnek

208 V, 10-HP, 4 kutuplu, 60 Hz, 4-bağlı asenkron motor tam yükte %5 kaymaya sahiptir.

- Bu motorun senkron hızı nedir?
- Nominal yükte bu motorun rotor hızı nedir?
- Nominal yükte bu motorun rotor frekansı nedir?
- Nominal yükte bu motorun mil momenti nedir?

$$n_{\text{sync}} = \frac{120 f_e}{p} = \frac{120 \cdot 60}{4} = 1800 \text{ rpm}$$

$$n_m = (1-s) n_{\text{sync}} = (1-0.05) 1800 = 1710 \text{ rpm} \quad \rightarrow \frac{r}{\text{min}}$$

$$f_r = s f_e = 0.05 \times 60 = 3 \text{ Hz}$$

$$T_{\text{load}} = \frac{P_{\text{out}}}{\omega_m} = \frac{10 \text{ hp} \cdot 746 \frac{\text{W}}{\text{hp}}}{1710 \frac{r}{\text{min}} \cdot 2\pi \frac{\text{rad}}{r} \cdot \frac{1 \text{ min}}{60 \text{ s}}} = 41.7 \text{ N.m}$$

A 400V, 6-pole, 3-phase, 50Hz, star connected induction motor gives the following measurements

$$V_{NL_no\ load} = 400V \quad I_{NL} = 7.5A \quad P_{NL} = 700W$$

$$V_{BR} = 150V \quad I_{BR} = 35A \quad P_{BR} = 4000W$$

Blocked rotor

$$R_s \text{ per phase is } 0.55\Omega \quad \frac{X_{ls}}{X_{lr}'} = 2$$

When the motor is operating at a slip of 4%, calculate

- stator current
- power factor
- net mechanical power output and torque
- efficiency

$$\text{From NL test } P_{rot} = 700 - 3 \cdot 7.5^2 \cdot 0.55 = 607W$$

$$R_{wfs} = \frac{(400/\sqrt{3})^2}{607/3} = 264\Omega \quad Z_{NL} = \frac{400/\sqrt{3}}{7.5} = 30.8\Omega$$

core winding
friction loss

$$X_m = \frac{1}{\frac{1}{30.8} - \frac{1}{264}} = 31\Omega$$

From BR test

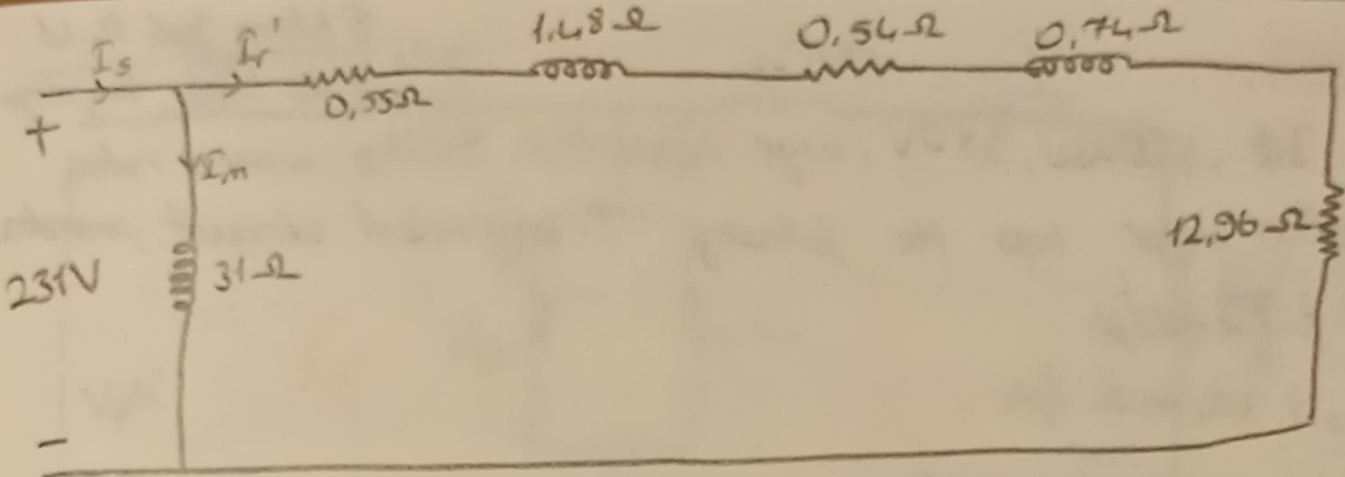
$$R_s + R_r' = \frac{4000/3}{35^2} = 1.09\Omega$$

$$R_s = 0.55\Omega \quad R_r' = 0.54\Omega$$

$$X_{ls} + X_{lr}' = \sqrt{\left(\frac{150/\sqrt{3}}{35}\right)^2 - (1.09)^2} = 2.22\Omega$$

$$\text{for } s = 0.04$$

$$X_{ls} = 1.48\Omega \quad X_{lr}' = 0.74\Omega$$



$$a) I_s = I_r' + I_m \quad I_r' = \frac{231}{14.05 + j2.22} = 16.24 \angle -9^\circ \text{ A}$$

$$I_s = 18.91 \angle -31.9^\circ \text{ A} \quad I_m = \frac{231}{j31} = 7.45 \angle -90^\circ$$

$$b) \text{pf} = \cos 31.9^\circ = 0.85 \text{ lagging}$$

$$c) P_{mi} = 3(I_r')^2 \frac{R_r'(1-s)}{s} = 3 \times 16.24^2 \times 12.96 = 10.25 \text{ kW}$$

$$P_{mnet} = 10.25 - P_{rot} = 10.25 - 0.607 = 9.64 \text{ kW}$$

$$\text{Since } N_r = (1-s) N_s, \quad N_r = 960 \text{ rpm}$$

$$T_{mnet} = \frac{P_{mnet}}{\omega_m} = \frac{9.64 \times 10^3}{2 \times \pi \times 960 / 60} = 96 \text{ Nm}$$

$$d) \eta = \frac{P_{out}}{P_{in}} \times 100 \quad P_{in} = \sqrt{3} V_s \times I_s \times \text{pf} = \sqrt{3} \times 231 \times 18.9 \times 0.85 = 11.13 \text{ kW}$$

$$\eta = \frac{9.64}{11.13} = 86.61 \%$$

Example

8 Nison Solu 16.61
Yildiz Saraylı Kocaeli

A 3 ϕ , 500kW, 380V, wye connected, 50Hz wound rotor induction motor has the following T-equivalent circuit parameters.

$$R_1 = 1.75 \text{ m}\Omega/\text{ph}$$

$$R_2' = 1.344 \text{ m}\Omega/\text{ph}$$

$$X_{1\sigma} = 15.7065 \text{ m}\Omega/\text{ph}$$

$$X_{2\sigma}' = 15.7065 \text{ m}\Omega/\text{ph}$$

$$R_{sc} = 10.027 \Omega/\text{ph}$$

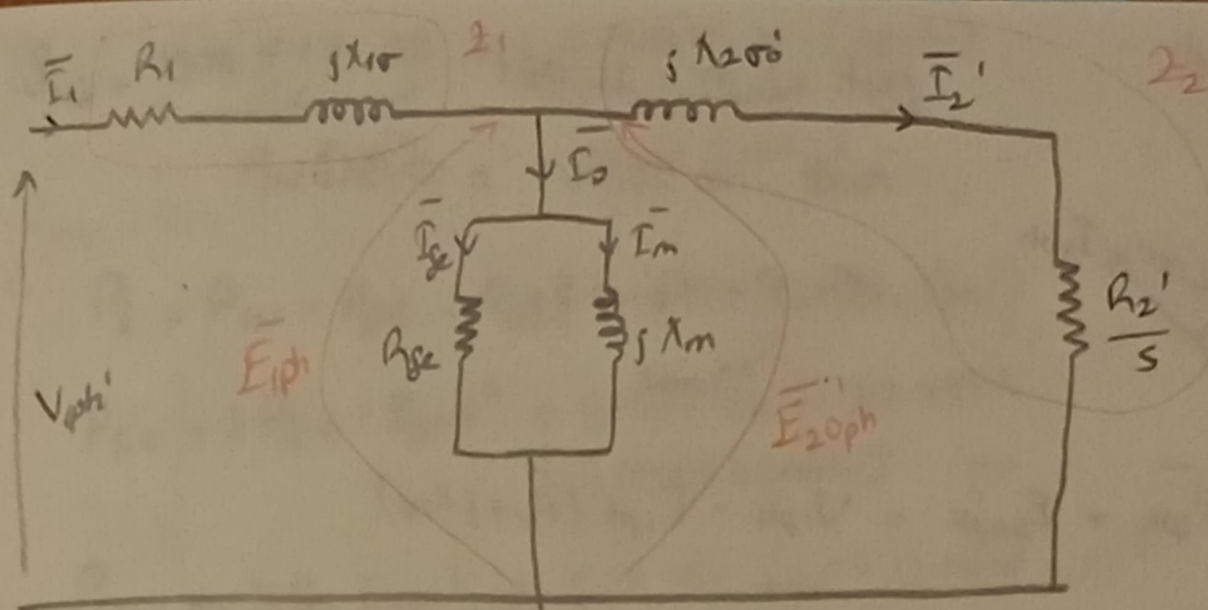
$$X_m = 0.643 \Omega/\text{ph}$$

$$P_{\text{gap}} = 9500 \text{ W}$$

Calculate the followings when
rated speed is equal to
 $n_r = 991 \text{ rpm}$ using T-equivalent circuit.

- Input current $I_{1\text{ph}}$, input power factor, $\cos \phi$,
- No-load branch current I_0
- Referred rotor current I_2'
- Input apparent (S_n), active (P_n) and reactive Q_n
- Internal (air-gap) power (P_i), internal mechanical power (P_{im}) and mechanical shaft power (P_m)
- Induced electromagnetic torque (T_i) and shaft torque (T_m)
- Efficiency of the motor.

T-Equivalent Circuit of an Induction Motor:



$$n_r = 991 \text{ rpm} \quad n_s = \frac{60f}{p} = \frac{120f}{2p} = \frac{60 \cdot 50}{p} \rightarrow \text{must be around } 1000$$

$$s = \frac{1000 - 991}{1000} = 0.009 \text{ (keynote or)}$$

$$p = 3$$

$$2p = 6$$

$$n_s = 1000 \text{ rpm}$$

$$\bar{V}_{1ph} = \frac{380 \angle 0^\circ}{\sqrt{3}} = 219.393 \angle 0^\circ [\text{V}]$$

$$\bar{Z}_{eq} = \bar{Z}_1 + (\bar{Z}_2 \parallel R_{se} \parallel jX_m)$$

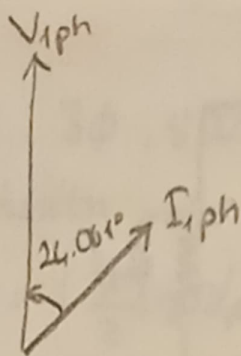
$$\bar{Z}_2' = \frac{1.314 \times 10^{-3}}{0.009} + j 15.7065 \times 10^{-3} [\Omega]$$

$$\bar{Z}_1 = 1.75 \times 10^{-3} + j 15.7065 \times 10^{-3} [\Omega]$$

$$\bar{Z}_{eq} = 0.1328 + j 0.0593 [\Omega] = 0.1454 \angle 24.069^\circ [\Omega]$$

$$\bar{I}_{1ph} = \frac{\bar{V}_{1ph}}{\bar{Z}_{eq}} = \frac{219.393 \angle 0^\circ}{0.1454 \angle 24.069^\circ} = 1508.468 \angle -24.061^\circ$$

$$= 1377.3126 - j 615.2133$$



$$\angle V_{ph} - \angle I_{ph} = 24.061^\circ$$

$$\cos \phi = \cos 24.061 = 0.913$$

$$S = P + jQ \rightarrow S \sin \phi$$

$$\rightarrow S \cos \phi$$

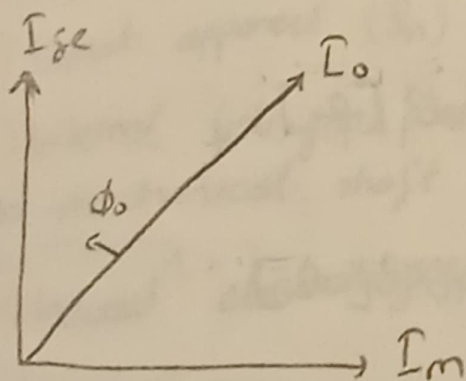
b) $\bar{E}_{1ph} = \bar{E}_{20ph} = \bar{V}_{1ph} - \bar{I}_{1ph} (\bar{R}_1 + jX_{1\sigma})$

$$\bar{E}_{1ph} = 207.32 - 20.56j = 208.34 \angle -5.66^\circ [V]$$

$$\bar{I}_{ge} = \frac{\bar{E}_{1ph}}{R_{ge}} = 20.78 \angle -5.66 [A]$$

$$\bar{I}_m = \frac{\bar{E}_{1ph}}{jX_m} = 324 \angle -95.66^\circ [A]$$

$$\bar{I}_0 = \bar{I}_{ge} + \bar{I}_m \quad \bar{I}_0 = 324.67 \angle -91.99^\circ [A]$$



c) $\bar{I}'_{2ph} = \bar{I}_{1ph} - \bar{I}_0 = 1418.72 \angle -11.83 [A]$

d) $S_{IN} = 3 \bar{V}_{1ph} \cdot \bar{I}_{1ph}^* = \sqrt{3} \bar{V}_{1pp} \cdot \bar{I}_{1pp}^*$

$$= 3219.993 \angle 0^\circ \cdot 1508.468 \angle 24.07^\circ$$

$$S_{IN} = \underbrace{906519,29}_{P_{IN} [W]} + j \underbrace{604917,2}_{Q_{IN} [VAR]} [VA]$$

$$e) P_i = P_{IN} - P_{ge} - P_{cu1} = 881586,94 [W]$$

$$P_{ge} = 3 R_{fe} |I_{fe}|^2 = 3 \cdot 10,027 (20,78)^2 = 12986,1 [W]$$

$$P_{cu1} = 3 R_1 I_{1N}^2 = 3 (1,75 \times 10^{-3}) (1508,67)^2 = 11946,25 [W]$$

$$P_{ie} = P_{cu2} = s \cdot P_i = s \frac{3 R_2' I_2'^2}{s} = 3 R_2' I_2'^2$$

$$P_{ie} = 7934,28 [W]$$

$$P_{im} = P_i - P_{ie} = (1-s) P_i = \frac{3(1-s)}{s} R_2' I_2'^2$$

$$P_{im} = 873652,65 [W]$$

$$P_m = P_{im} - P_{gsw} = 864152,65 [W]$$

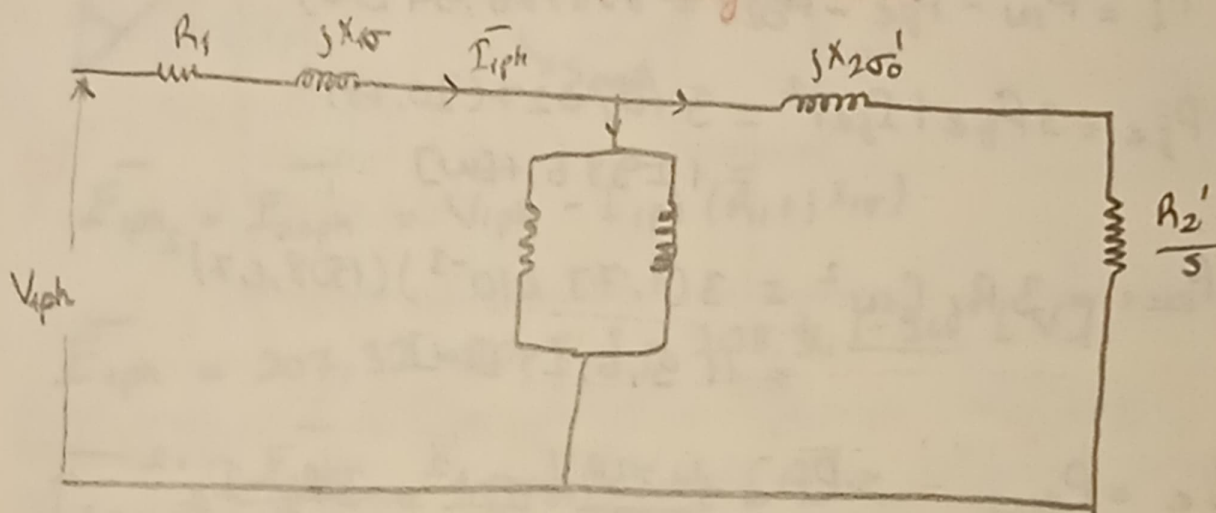
$$f) T_i = \frac{P_i}{\omega_s} = \frac{P_{im}}{\omega_r} = \frac{873652,65}{\frac{2\pi \cdot 991}{60}} = 8418,54 [Nm]$$

$$\text{Shaft torque } T_m = \frac{P_m}{\omega_r}$$

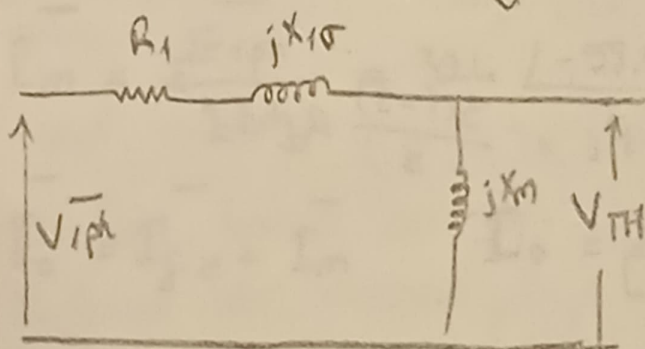
$$T_m = \frac{864152,65}{\frac{2\pi \cdot 991}{60}} = 8326,99 [Nm]$$

$$8) \eta = \% \frac{P_m}{P_{in}} \times 100 = \frac{86413265}{90651929} \times 100 = 95.33\%$$

Pull-out (Torque, Speed, Slip) Using Thevenin



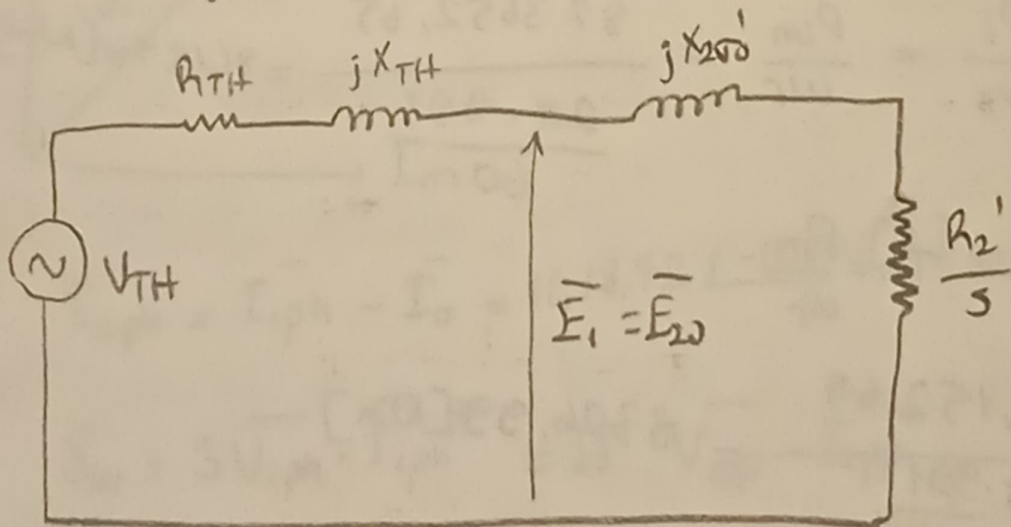
↓ Reduce the circuit, ignore R_2



$$V_{TH} = \frac{jX_m}{R_1 + jX_{1\sigma} + jX_m} \cdot \bar{V}_{1ph}$$

$$|V_{TH}| = \frac{X_m}{\sqrt{R_1^2 + (X_{1\sigma} + X_m)^2}} \cdot |V_{1ph}|$$

$$Z_{TH} = \frac{jX_m (R_1 + jX_{1\sigma})}{jX_m + R_1 + jX_{1\sigma}} = R_{TH} + jX_{TH}$$



Since $X_m \gg X_{1\sigma}$ and $X_{1\sigma} + X_m \gg R_1$

$$R_{TH} \approx R_1 \left(\frac{X_m}{X_{1\sigma} + X_m} \right)^2 \quad X_{TH} \approx X_{1\sigma}$$

$$\bar{I}_1 = \bar{I}_2' = \frac{\bar{V}_{TH}}{R_{TH} + \frac{R_2'}{s} + j(X_{TH} + X_{2\sigma_0}')}$$

$$|\bar{I}_2'| = \frac{|\bar{V}_{TH}|}{\sqrt{\left(R_{TH} + \frac{R_2'}{s}\right)^2 + (X_{TH} + X_{2\sigma_0}')^2}}$$

$$P_i = 3 \cdot I_2'^2 \cdot \frac{R_2'}{s} \quad T_i = \frac{3}{\omega_s} \cdot I_2'^2 \cdot \frac{R_2'}{s}$$

$$s_{pull-out} = \frac{R_2}{\sqrt{R_{TH}^2 + (X_{TH} + X_{2\sigma_0}')^2}}$$

$$T_{pull-out} = \frac{3}{\omega_s} \frac{V_{TH}^2}{2[R_{TH} + \sqrt{R_{TH}^2 + (X_{TH} + X_{2\sigma_0}')^2}]}$$

Example

A 460V, 25 HP, 60 Hz, four pole, Y connected wound motor has the following impedances in Ω/ph referred to stator circuit.

$$R_1 = 0.641 \Omega/ph$$

$$R_2 = 0.332 \Omega/ph$$

$$X_m = 26.3 \Omega/ph$$

$$X_{1\sigma} = 1.1 \Omega/ph$$

$$1 \text{ HP} = 746 \text{ W}$$

$$X_{2\sigma} = 0.464 \Omega/ph$$

a) What is the maximum (pull-out) torque of this motor? At what slip and speed does it occur?

b) What is the starting torque of this motor?

c) When the rotor resistance is doubled, what is the speed which maximum torque occurs?

d) What is the new starting torque?

e) Draw the $T-s$ characteristic of the motor before and after conditions given above on the same graph.

$$V_{TH} = V_{1ph} \cdot \frac{X_m}{\sqrt{R_1^2 + (X_{1\sigma} + X_m)^2}} = 255,1 \text{ V}$$

$$V_{1ph} = \frac{460}{\sqrt{3}} = 266 \text{ V} \quad R_{TH} \approx R_1 \left(\frac{X_m}{X_{1\sigma} + X_m} \right)^2 = 0,59 \Omega/ph$$

$$X_{TH} \approx X_{1\sigma} = 1,106 \Omega/ph$$

a)

$$s_{pull-out} = \frac{R_2}{\sqrt{R_{TH}^2 + (X_{TH} + X_{2\sigma}')^2}} = 0,198$$

$$n_s = \frac{60 \cdot 60}{2} = 1800 \text{ rpm}$$

$$n_{pull-out} = n_s (1 - s_{pull-out}) = 1444 \text{ rpm}$$

$$T_{pull-out} = \frac{3}{2\omega_s} = \frac{V_{TH}^2}{R_{TH} + \sqrt{R_{TH}^2 + (X_{TH} + X_{2\sigma}')^2}} = 229 \text{ Nm}$$

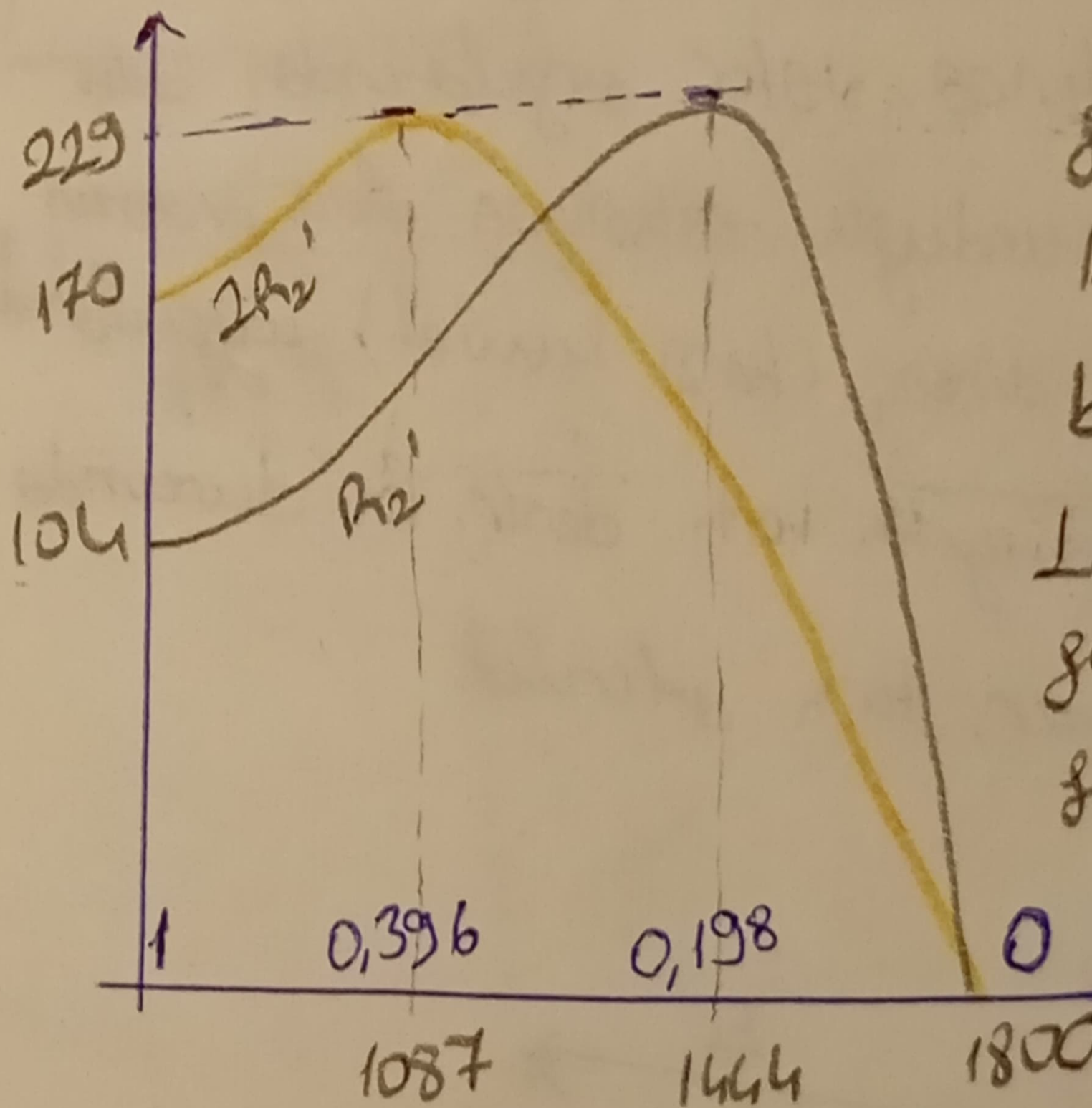
$$\left. \begin{aligned}
 s_{p-out_1} &= 0.198 \\
 n_{p-out_1} &= 1444 \text{ rpm} \\
 T_{p-out_1} &= 229 \text{ Nm} \\
 T_{starting_1} &= 104 \text{ Nm}
 \end{aligned} \right\} \begin{aligned} &\text{After} \\ &\text{doubling} \\ &R_2' \end{aligned}$$

$$\begin{aligned}
 s_{pull-out_2} &= 0.396 \\
 n_{pull-out_2} &= 1087 \text{ rpm} \\
 T_{pull-out_2} &= 229 \text{ Nm} \\
 T_{starting_2} &= 170 \text{ Nm}
 \end{aligned}$$

b) By setting $s=1$

$$T_L = \frac{3V_{TH}^2 R_2}{\omega_s [(R_{TH} + R_2')^2 + (X_{TH} + X_{200}')^2]} = 104 \text{ Nm}$$

c)



T_i is the induced electromagnetic torque, if there were no mechanical losses, rotor would have this.

$$T_i = \frac{P_{im}}{2\pi nr/60}$$

T_m is the torque that actually performs work. After accounting for losses like friction, windage etc.

$$T_m = \frac{P_m}{2\pi nr/60}$$

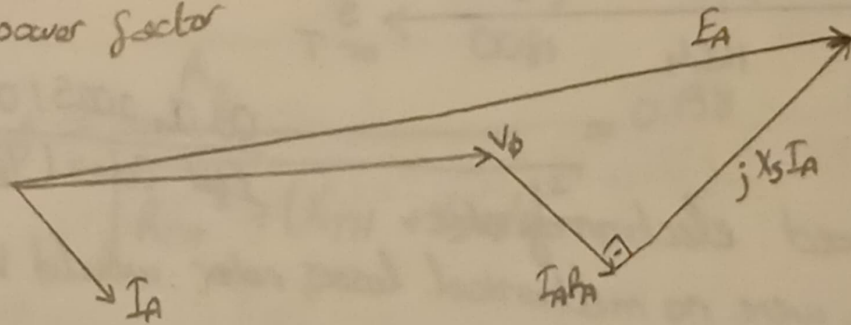
$$\eta = \% \frac{P_m}{P_{in}}$$

Permanent magnets, dışardan elektrik verilmeden sürekli manyetik alan üreten maddelerdir. Bu özellikli motor ve generatörlerde alan kaynağı olarak kullanılırlar. Bu alana yerleştirilen bir iletken tel hareket ederse elektrik üretir. Telden geçen akım manyetik alan girerse hareket oluşur. (Generator-motor) Bunu ^{B-yıld} permanent magnet dışında elektromagnetik (bobin+akım) ile oluşturabiliriz.

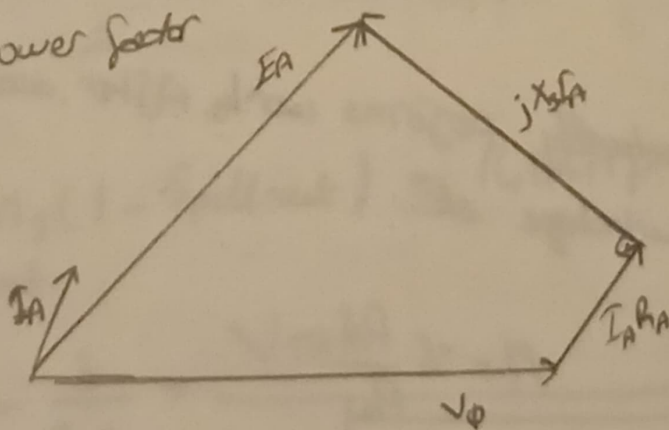
Senkron generatörlerde rotor dışardan senkron hızda döndürülür, manyetik alan hazır, stator bu değişen alanı göreyak gerilim üretir. Rotorun dönme hızı şebeke frekansına kilitlenir. 50 Hz. Elektrik yükü artınca, stator sargılarından akım akır. Bu akımın oluşturduğu reaksiyon alanı, rotorun alanına karşı bir elektromanyetik direnç (karşı kuvvet) uygular. Rotorun dönmesi zorlaşır. Elektromanyetik tork derir. Bu durumda türbine daha fazla enerji yollarız, tork artırılır.

Phasor Diagrams

Lagging power factor



Leading power factor



Ornek

A four-pole 3 ϕ synchronous generator has 0.8 pf lagging

$$S_N = 400 \text{ kVA}$$

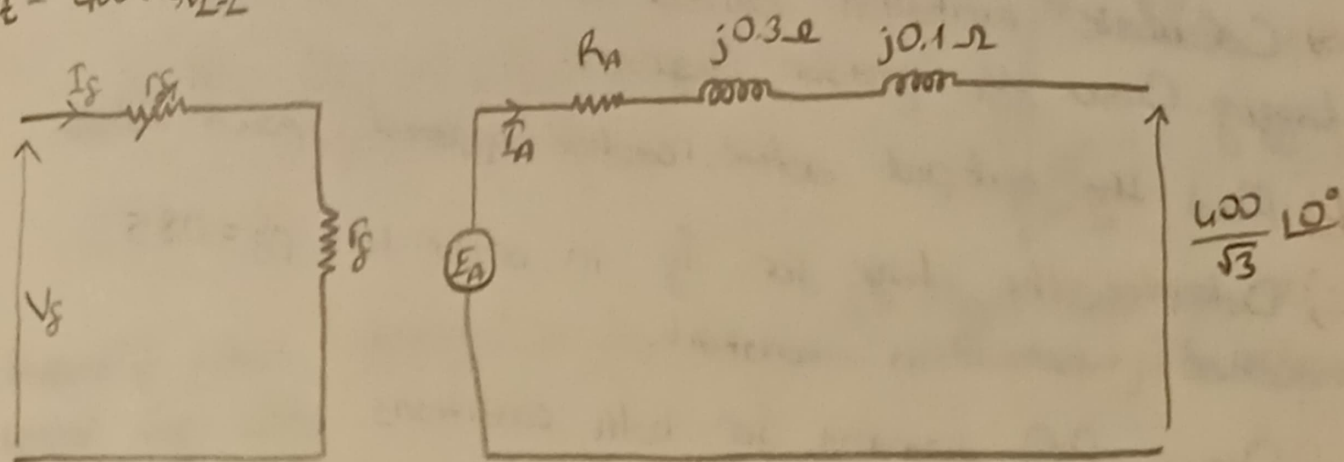
$$V_L = 400 \text{ V} \rightarrow V_{L-L}$$

$$P_{\text{gsw}} + P_{\text{ge}} = 15 \text{ kW}$$

$$R_A = 8 \text{ m}\Omega$$

$$X_A = 0.3 \Omega$$

$$X_\sigma = 0.1 \Omega$$



$$|I_A| = \frac{S_N}{\sqrt{3} \cdot V_L} = 577 \frac{\text{A}}{\cos^{-1}(0.8)}$$

$$\begin{aligned} E_g &= V_\phi + I_A R_A + I_A j X_s \\ &= 231 \angle 0^\circ + 577.39 \angle -36.87^\circ (8 \times 10^{-3} + j0.4) \\ &= 415.2 \angle 25.99^\circ \rightarrow \delta = 25.99^\circ \end{aligned}$$

$$\psi = \phi + \delta = 62.86^\circ$$

$$P_i = 3 \frac{|E_g| |V_\phi|}{|X_s|} \sin \delta$$

$$P_i = 3 \frac{415.2 \times 231}{0.4} \sin(25.99^\circ)$$

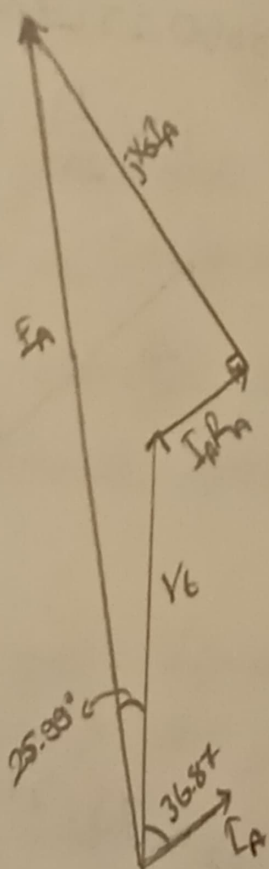
$$P_i = 315.14 \text{ kW}$$

$$P_{cu} = 3 I_\phi^2 R_g = 8 \text{ kW}$$

$$P_{\text{out}} = P_i - P_{cu} = 307.14 \text{ kW}$$

$$P_{\text{in}} = P_i + P_{\text{gsw}} + P_{\text{ge}} = 330.14 \text{ kW}$$

$$\eta = \frac{307.14}{330.14} = 93\%$$



Example

3 ϕ wye connected 187.5 MVA, 15 kV, 50 Hz round rotor synchronous generator has a reactance 2.57 Ω /ph $R_s \approx 0$

$$E_g = 250 I_g$$

- Calculate excitation current at 75% load at $\cos \phi = 0.75$ lagging. Draw the phasor diagram.
- Find the output active, reactive, apparent power values.
- Determine the change for I_g in order to $\text{pf} = 0.85$ inductive when P is constant.
- Draw PQ diagrams for both conditions.
- Find new Q_{out} , P_{out} , S_{out} .

$$I_{\text{aw}} = \frac{187.5 \times 10^6}{\sqrt{3} \times 15000} = 7216.88 \text{ A}$$

$$|I_A| = 0.75 \times 7216.88 = 5412.66 \text{ A}$$
$$\cos \phi = 0.75 \rightarrow \phi = 41.41^\circ$$

$$\bar{I}_A = 5412.66 \angle -41.41^\circ \quad V_\phi = \frac{15}{\sqrt{3}} \text{ kV} = 8660.25 \angle 0^\circ \text{ V (reference)}$$

$$\bar{E}_g = V_\phi + j X_s \bar{I}_A = 20679.66 \angle 30.3^\circ \quad I_g = \frac{E_g}{250} = 82.72 \text{ A}$$

$$P_1 = 3 \cdot 8660.25 \cdot 5412.66 \cdot \cos(41.41^\circ)$$
$$= 105.47 \text{ MW}$$

$$Q_1 = 92.953 \text{ MVAR}$$

$$P_2 = 3 \cdot V_\phi \cdot I_{A2} \cdot \frac{0.85}{\cos \phi_2} = 105.47 \text{ MW}$$

$$I_{A2} = 6773.89 \angle -31.79^\circ$$

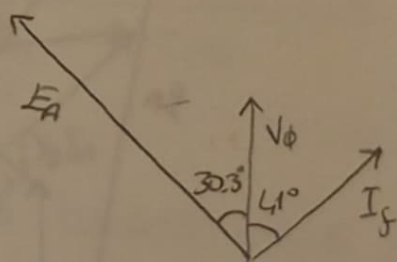
$$E_{g2} = V_\phi + I_{A2} j X_s = 18378 \angle 34.59^\circ$$

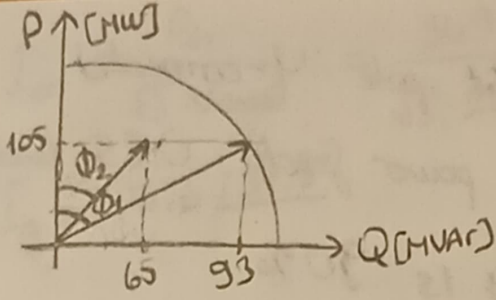
$$I_{g2} = 73.51 \text{ A}$$

$$Q_2 = P_2 \cdot \tan(31.79^\circ) = 65.39 \text{ MVAR}$$

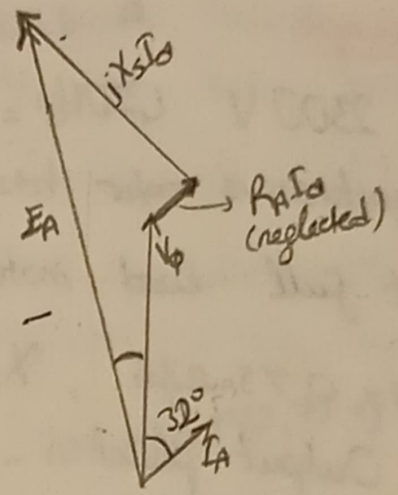
$$S_2 = 124.08 \text{ MVA}$$

$$\frac{I_{g2}}{I_{g1}} = 88.8\%$$

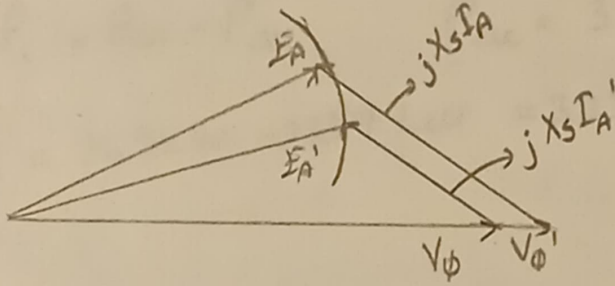




$$\tan \psi = \frac{V_{\phi} \sin \Phi + I_A X_s}{V_{\phi} \cos \Phi + I_A R_A}$$



25 Mayıs Pazartesi 11.17
İkapsizlik yapan sağladığı Q I_A vektörünü dindirek jI X_s vektörünü aşağı çeker.



Yeni $V_{\phi}' > V_{\phi}$
leading'e özel bir durum.
İkapsizlik etkisi uç gerilimi artırır.

Senkron generatörde,
Rotor döner → Statorda gerilim indüklenir.
Stator yük çekmeye başlarsa → Statordan akım çıkar
Bu akım, rotorun oluyula etkileşir → Rotoru yavaşlatan bir moment doğur.
indüklenen zıt moment
Generatörü sabit hızda döndürmek için dışardan bir mekanik moment (mül gücü) uygulanmalıdır.
Yük artarsa → akım artar → zıt moment artar → daha fazla mekanik moment gerekir.

Example

A 2300 V 420 Hp 50 Hz eight pole 4-connected synchronous motor has a rated power factor 0.85 leading.

At full load motor efficiency is 90%

$R_s = 0.75 \Omega/\text{ph}$, $X_s = 12 \Omega/\text{ph}$ Find the;

- Output power
- Input power
- n_r mechanical speed
- I_s armature current
- E_s induced voltage
- P_i (air gap power)
- Motor losses

a) $n_s = \frac{120 f}{p} = 750 \text{ rpm}$ $1 \text{ HP} = 0.746 \text{ kW}$

$$P_{\text{out}} = 420 \text{ Hp} \cdot 0.746 = 314 \text{ kW}$$

$$T_{\text{out}} = \frac{P_{\text{out}}}{\omega_r} = \frac{314}{\frac{2\pi \cdot 750}{60}} = 3990 \text{ Nm}$$

b) $P_{\text{in}} = \frac{P_{\text{out}}}{\eta} = \frac{314}{0.9} \approx 349 \text{ kW}$

c) $n_s = n_r = 750 \text{ rpm}$

$$d) \quad I_a = \frac{P_{in}}{\sqrt{3} V \cos \theta} = \frac{349 \text{ k}}{\sqrt{3} \cdot 2300 \cdot 0,85} = 103,6 \text{ A}$$

$$\phi = 31,8^\circ$$

$$I_A = 103,6 \angle 31,8^\circ$$

$$e) \quad V_T = \frac{2300 \angle 0^\circ}{\sqrt{3}} = 1328 \angle 0^\circ \text{ V}$$

$$E_g = \bar{V}_1 - I_A (R_s + jX_s) = 132,8 \angle 0^\circ - 103,6 \angle 31,8^\circ (0,75 + j12)$$

$$= 1313,65 \angle -56,66^\circ \text{ V}$$

$$f) \quad P_i = P_{in} - P_{cu} \quad P_{cu} = 3 I_a^2 R_s = 3 \cdot 103,6^2 \cdot 0,75 = 23870$$

$$P_i = 349 \text{ kW} - 23,87 \text{ kW} = 325 \text{ kW}$$

$$g) \quad P_{out} + P_{core} + P_{sw} = P_i$$

$$314 + \underbrace{P_{core} + P_{sw}}_{11 \text{ kW}} = 325 \text{ [kW]}$$

$$P_{cu} = 23,87 \text{ kW} \quad P_{total \text{ loss}} = 34,87 \text{ kW}$$

Example

A three phase Y-connected, round rotor synchronous generator has following parameters:

$$S_n = 400 \text{ MVA} \quad V_T = 400 \text{ V} \quad R_s = 8 \text{ m}\Omega / \text{ph} \quad X_s = 0,4 \Omega / \text{ph}$$

$$P_{sw} + P_{core} = 15 \text{ kW}$$

a) Calculate internal generated voltage E_f , load angle (δ) and internal power angle at 0.8 pf lagging with full load.

b) Draw the phasor diagram.

c) Find the input internal power, output power, reactive power and efficiency of the generator

d) If I_f excitation current increased 15%. Calculate the new value of reactive power at same load conditions.

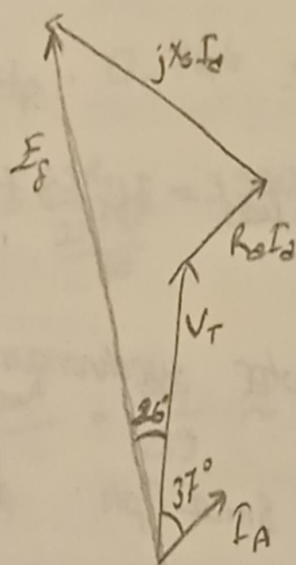
$$|I_A| = \frac{|S_n|}{\sqrt{3} |V_T|} = \frac{400 \text{ kVA}}{\sqrt{3} \cdot 400} = 577.35 \text{ [A]}$$

$$X_s = 0.4 \Omega \quad V_\phi = \frac{400}{\sqrt{3}} = 230.94 \angle 0^\circ \text{ [V]}$$

$$\cos \phi = 0.8 \quad \phi = -36.87^\circ \text{ lagging} \quad I_s = 577.35 \angle -36.87^\circ$$

$$E_f = V_\phi + I_s (R_s + jX_s) = 230.94 \angle 0^\circ + 577.35 \angle -36.87^\circ (0.08 + j0.4)$$

$$\bar{E}_f = 415.2 \angle 25.99^\circ \quad \delta = 25.99^\circ \quad \psi = \phi + \delta = 62.86^\circ$$



$$P_{out} = S_n \cdot \cos \phi = 320 \text{ kW}$$

$$P_{cu} = 3 I_s^2 R_s = 8 \text{ kW}$$

$$P_{sw} + P_{core} = 15 \text{ kW}$$

$$P_{in} = P_{sw} + P_{core} + P_{cu} + P_{out} = 343 \text{ kW}$$

$$\frac{320}{343} \times 100 = 93.2\%$$

$$Q = \sqrt{3} V_T I_s \sin \phi = \sqrt{3} \cdot 400 \cdot 577.35 \sin(36.87^\circ)$$

$$Q = 240 \text{ kVAR}$$

$E_f = f(I_f)$ Linear $\rightarrow$ saturation neglected

$$\bar{E}_f' = 1.15 \bar{E}_f \quad |\bar{E}_f'| = 477.48 \text{ V}$$

$$\frac{3|\bar{E}_f'| |V_T| \sin \delta'}{X_s} = \frac{3|\bar{E}_f| |V_T| \sin \delta}{X_s}$$

$$\sin \delta' = \frac{\sin \delta}{1.15} \quad \delta' = 22.8^\circ$$

$$\bar{E}_f' = \bar{V}_T + \bar{I}_A (R_s + jX_s)$$

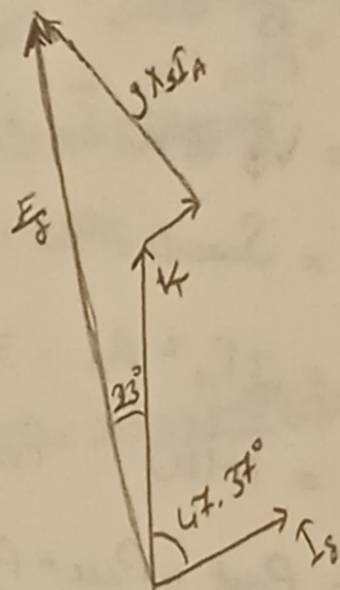
$$477.48 \angle 22.8^\circ = 230.94 \angle 0^\circ + I_A (0.008 + j0.4)$$

$$\bar{I}_A = 698.082 \angle -47.37^\circ \text{ A}$$

$$Q = \sqrt{3} V \bar{I}_A \sin \phi$$

$$= \sqrt{3} \cdot 400 \cdot 698 \sin(47.37^\circ)$$

$$= 355.7 \text{ kVAR}$$



Example

A 13.8 kV 50 MVA 0.9 pf lagging

Y-connected, 60 Hz, synchronous generator has

$X_s = 2.5 \Omega/\text{ph}$ and $R_s = 0.2 \Omega/\text{ph}$ $P_{\text{core}} = 1.5 \text{ MW}$
 $P_{\text{sw}} = 1 \text{ MW}$

a) What is the efficiency of generator at rated load and what is the induced voltage at given condition of 0.9 pf lagging?

- b) What is the induced voltage at 0.9 pf leading?
- c) Draw the phasor diagrams for a and b
- d) If input mechanical power of the generator increased 5% at the same apparent power. Calculate load angle and power angles of generator at lagging pf.

$$I_a = \frac{S_n}{\sqrt{3} V_L} = \frac{50 \text{ M}}{\sqrt{3} \cdot 13800} = 2092 \text{ A} \quad \phi = -25.8^\circ$$

$$V_\phi = \frac{V_L}{\sqrt{3}} = 7967 \angle 0^\circ \text{ V}$$

$$\bar{E}_f = \bar{V}_\phi + R_A \bar{I}_A + jX_s \bar{I}_A = 11544 \angle 23.1^\circ$$

$$P_{out} = S \cos \phi = 50 \cdot 0.9 = 45 \text{ MW}$$

$$P_{cu} = 3 I_A^2 R_A = 3 \cdot 2092^2 \cdot 0.2 = 2.6 \text{ MW}$$

$$P_{sw} = 1 \text{ MW} \quad P_{core} = 1.5 \text{ MW}$$

$$P_{in} = P_{out} + P_{cu} + P_{sw} + P_{core} = 50.1 \text{ MW}$$

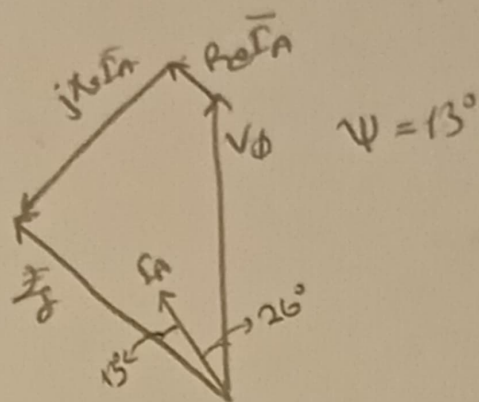
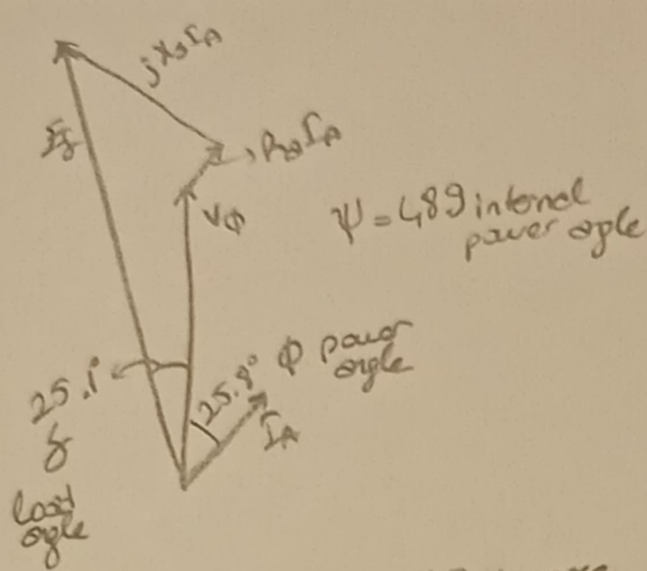
$$Q = S \sin \phi = 50 \sin(25.8^\circ) = 21.76 \text{ [MVAR]}$$

$$\eta = \frac{45}{50.1} = 89.8 \%$$

$$\text{If pf is leading } 0.9 \rightarrow \phi = 25.8^\circ$$

$$\bar{E}_f = 7967 \angle 0^\circ + 2092 \angle 25.8^\circ (0.2 + j2.5)$$

$$\bar{E}_f = 7793 \angle 38.8^\circ \quad \delta = 38.8^\circ$$



$\xrightarrow{\%S \text{ increase}}$

$P_{out} = 4.5 \text{ MW}$

$S = 50 \text{ MVA}$

$P_{out}' = 47.25 \text{ MW}$

$P_{out}' = 50 \cdot \cos \phi' \quad \cos \phi' = 0.945$
 $\phi' = 19^\circ$

new p.f. $\rightarrow 0.945$ lagging

$Q' = S \sin \phi = 50 \cdot \sin 19^\circ = 16.35 \text{ MVAr}$

$I_A' = 2092 \angle -19^\circ \text{ A}$

$\bar{E}_g' = \bar{V}_\phi + I_A (R_s + jX_s) = 7967 \angle 0^\circ + 2092 \angle -19^\circ (0.2 + j2.5)$

$E_g' = 11154 \angle 25.5^\circ \text{ V}$

$\delta = 25.5^\circ$
 $\phi = 19^\circ$
 $\psi = 44.5^\circ$

